Muhammad Rayyan Masroor

FULL-STACK DEVELOPER · MACHINE LEARNING · AI AUTOMATION

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Computer Science undergraduate at FAST NUCES who builds full-stack apps, machine learning projects and AI automation. Shipped client e-commerce platforms on Next.js and Supabase, a local AI lead generation pipeline, three live football ML apps and an offline voice translator that runs entirely on the phone.

TECHNICAL SKILLS

Languages	Python, TypeScript, JavaScript, C++, C, C#, Java, SQL
ML & Data	scikit-learn, XGBoost, pandas, NumPy, matplotlib, K-Means, PCA, cross validation, model calibration
AI & Automation	n8n, Make.com, Ollama, Playwright, Transformers.js, ONNX Runtime, Whisper, LLM/RAG workflows
Web	Next.js, React, Tailwind CSS, Vite, PWA / Service Workers, Node.js, Express, FastAPI, REST APIs
Databases	Supabase, PostgreSQL, MongoDB, MySQL, SQLite
Tools	Git/GitHub, Vercel, Cloudflare Pages, Docker, pytest

EXPERIENCE

Software Development Intern (Incoming) · *Inotech Solutions (Pvt.) Ltd.* — Starting Oct 2026

- Design and development of database and information systems, full-stack applications and AI/ML applications.

MIS Development Intern · *Government of Nigeria* — Jun 2026 to Present

- Building a Management Information System for a government agency's institutional records.
- Designed relational schemas and backend logic for data entry, storage and retrieval with role-based access, plus public and admin web interfaces for government reporting.

Teaching Assistant (OOP, Data Structures) & Lab Instructor (COAL) · *FAST NUCES, Islamabad* — Jan 2025 to Present

- Grade work, hold office hours and help debug C++ and x86 assembly for 500+ students across OOP, Data Structures and COAL.
- Wrote custom problem sets that raised average lab scores by 15%.

SELECTED PROJECTS

AI Lead Generation & Outreach Automation — Startup · 2026

- Built a local two-stage pipeline (Node.js/Playwright + Python/FastAPI/Ollama) that finds businesses on Google Maps and enriches leads at zero per-call LLM cost. n8n runs it hourly across rotating niches and cities.
- Built a full-stack CRM (Next.js, Supabase/PostgreSQL) as the system of record, and a 3-step outreach sequencer with daily caps, send gaps and a content-safety gate that logs every send and refusal.
- A local LLM scores each lead's website for AI and chatbot opportunities, then drafts a personalised first email.

Football Analytics Suite: 3 live ML apps — Personal · 2026

- **Match Outcome Predictor:** calibrated XGBoost and Random Forest on leakage-free, time-ordered features, with 52% accuracy on an unseen La Liga season vs 54% for bookmakers.
- **Transfer Value Predictor:** regression on 726 players' stats against Transfermarkt values, with 5-fold out-of-fold predictions (R^2 0.79) to flag mispriced players.
- **Player Similarity Scout:** cosine similarity on 14 per-90 stats, plus K-Means archetypes and a PCA style map across 800+ players.
- All three export Python/scikit-learn models as JSON, run in the browser on Vercel and retrain on a schedule.

Offline Voice Translator (EN ↔ RU) · ru-translator.pages.dev — Personal · 2026

- Installable PWA (React, TypeScript, Vite) that translates speech between English and Russian with no internet. Whisper, OPUS-MT and speech output all run on the device through Transformers.js and ONNX Runtime WASM.
- Built a hallucination gate for Whisper (silence check, blocklist, repetition detection), resumable model downloads stored in IndexedDB, and COOP/COEP headers for multi-threaded WASM.

Char Meem Clothing: Full-Stack E-Commerce — Client · 2026

- Production clothing store (Next.js, Supabase/PostgreSQL) with a Fabric.js design studio for 4 print areas, an admin panel, promo codes and Resend order emails.
- Checkout runs as a PL/pgSQL function that locks product variants, re-checks stock and recomputes prices on the server.

ACHIEVEMENTS

CodeForces 1300+ rating · LeetCode 170+ problems solved · ICPC Speed Programming participant

EDUCATION

BS Computer Science · *FAST NUCES, Islamabad* — Expected 2027

Relevant coursework: Data Structures & Algorithms, OOP, Operating Systems, AI, Database Systems, Software Design & Analysis, Web Programming, Parallel & Distributed Computing, High Performance Computing.